CMS TFPX Phase-II upgrade	Standard Operating Procedure	
	SOP No.:	DC
	Title:	**Dee construction: Gluing parts into Dee mechanical structure**
	Revision:	V3
	Date:	April 26, 2026

Abstract

Describes the procedures to glue co-cured Carbon fiber-foam halves housing PEEK inserts, and titanium flower pattern evaporator, using the robotic gantry. The procedure takes place in the clean room. This is a two-person procedure.

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I. Scope

This process describes the assembly of the D-shaped mechanical structure ("Dee") stage for the TFPX cartridge integration process.

II. Purpose

The "Dee" is the mechanical structure that supports the silicon modules. It also provides cooling and support for power cables and e-links.

III. Definitions

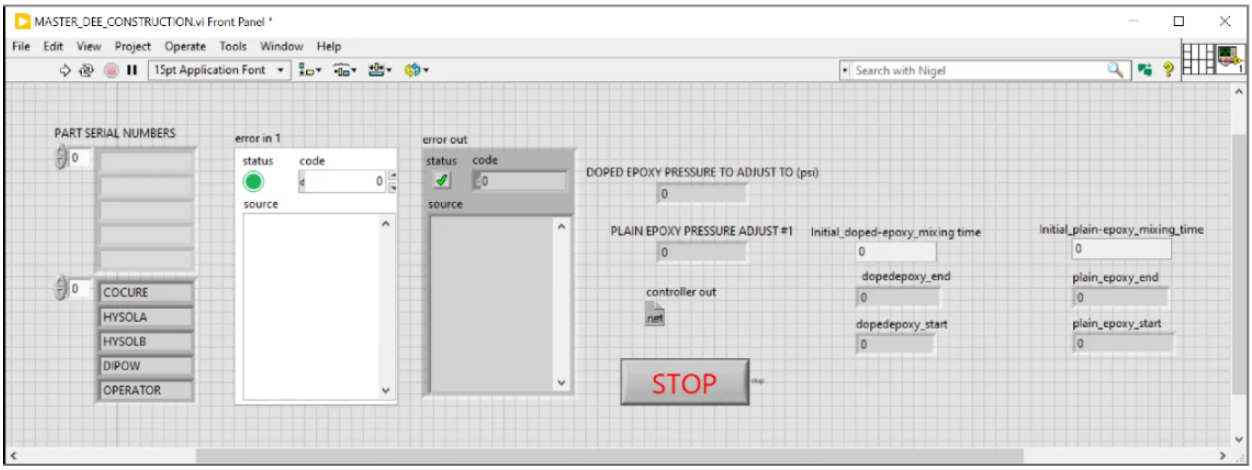
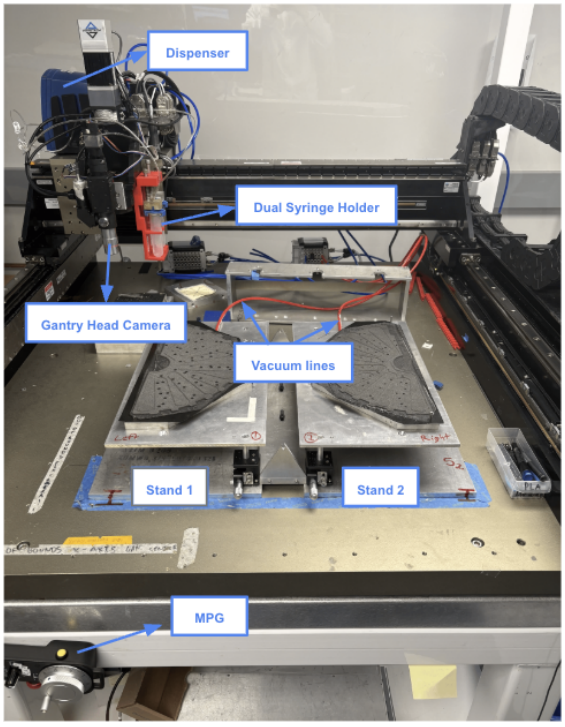


Figure 1: Front Panel of MASTER_DEE_CONSTRUCTION.vi

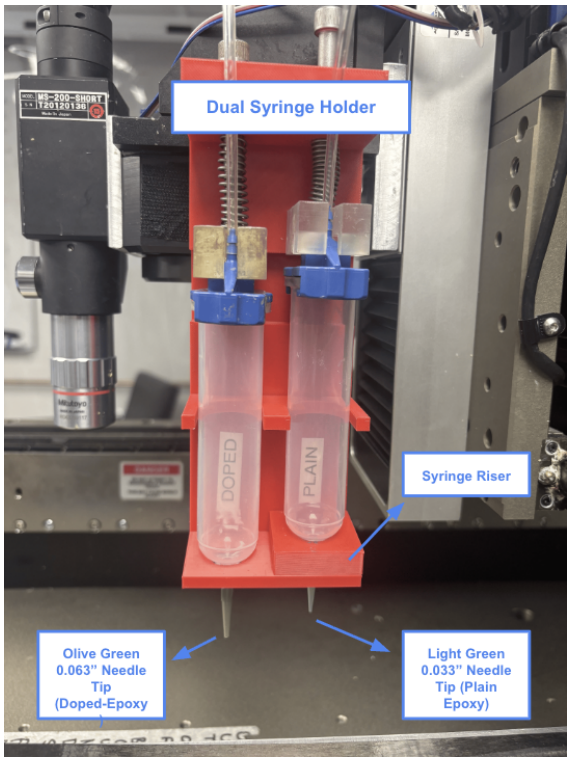
- Motion Composer: is a program that runs GCODE. GCODE is a set of instructions for the Gantry to move along. Think of a GCODE path like the path a paintbrush needs to be moved along to make a painting.
- Flower Pattern: The cooling tube or evaporator has a flower shape which we call here the flower pattern. The Gantry needs to deposit epoxy while it is moving along the flower pattern path in the Dee. For this, we have a GCODE script/program run by Motion composer.
- Carbon fiber-foam co-cure (CFF): The Dee structure is formed by two skins of carbon fiber and two bodies of carbon foam. One carbon fiber skin and carbon foam are co-cured in one entity, so that the Dee is composed of 2 halves of CFF (CFF_A and CFF_B). In between the two CFF, inserts and evaporator are sandwiched.
- Insert Filling: Dee has some PEEK inserts that will serve as clamping points for structures/objects on it. The Gantry moves to the location of the insert holes and deposit epoxy while STOPPED at that position.

- Front panel: It is the window through which the user interact with the LabVIEW control program. It shows statistical information, status of the vacuum system, tools and elements in use and other relevant information (see figure 1)

IV. Equipment



(a) Gantry setup. Labels indicate the main parts of the setup.



(b) Dual Syringe Holder. Labels indicate key attributes of the holder.

Figure 2: Primary Equipment and Setup for Dee Construction.

1. Gantry (check that everything (labeled in the figure 2a) is installed and working properly.
   - Syringe holder and riser piece (Figure 2b)
   - Stands with alignment screws
   - Pressurised air Dispenser - Ultimus II
   - Vacuum manifold and lines.
   - Gantry head camera with light
2. Parts and tools

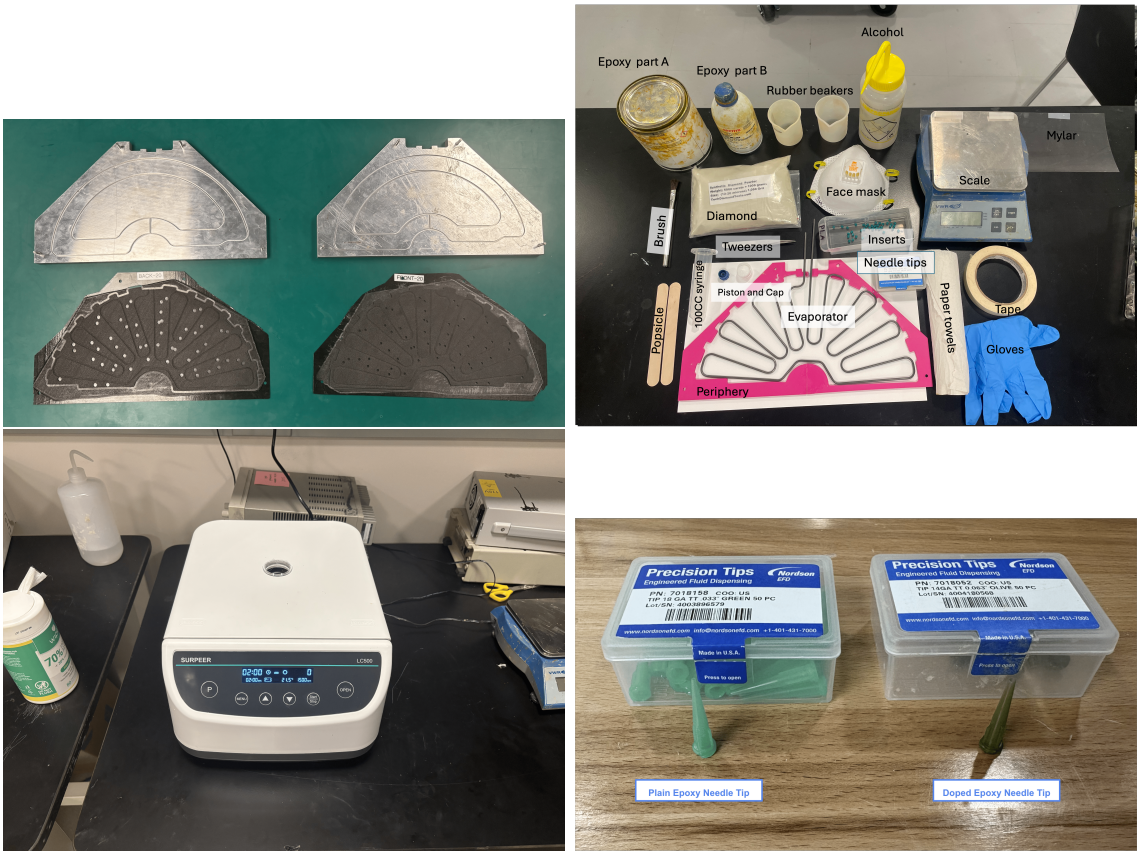


Figure 3: Parts and tools used in the DEE manufacturing process. Materials listed for epoxy preparation should be located on the table inside room PSB 375 where the epoxy mixing will be performed; the rest of materials must be located on the table in front of the gantry except for evaporator, inserts and tweezers which should be located on the table left to the gantry.

- Side A Half-Dee CFF_A
- Side B Half-Dee CFF_B
- PEEK Periphery
- EA9394 Epoxy parts A and B.
- Flower-Pattern Shape Titanium CO_2 tube/evaporator
- Box with PEEK inserts (56 will be used)
- Tweezers
- Two bottom jig plates (with vacuum lines)
- One top jig plate
- Both 3D printed tools for sealing tubing and inserts
- Flower pattern cover piece (for spreading epoxy)

- Small scooping tools for epoxy removal from holes
- Neodymium magnets for securing CFF Half-Dee to Jig plate

3. Epoxy preparation **(Performed in PSB 375)**

- Bag of 12-20 Micron Diamond Powder
- EA9394 Epoxy Part A
- EA9394 Epoxy Part B
- 2 Popsicle Sticks
- 2 Large Syringes, 2 Pistons, 2 Blue stopper caps
- 0.063" Olive green Nordson fluid dispensing tip
- 0.033" Light green Nordson fluid dispensing tip
- Digital Scale
- Large Centrifuge with Counterweights Installed
- Paper Towels
- Isopropyl Alcohol
- Rubber beakers
- Silicone Gloves
- N95 Mask
- Extension cord
- Square plastic sheet
- Stopwatch
- Box Fan
- Mylar Plastic sheet
- Painter's Tape
- Small soft paint brush

V. Procedure

Person 1:

1. Check that the NPaq, power supplies, dispenser and vacuum pumps are turned on. Turn on any if necessary

2. Check compressed air lines are open

3. Open LabView control software `C:\Users\co2_clt\Documents\MASTER_DEE_Construction\MASTER_DEE_CONSTRUCTION.vi` Ensure that it opens in Labview 2025.

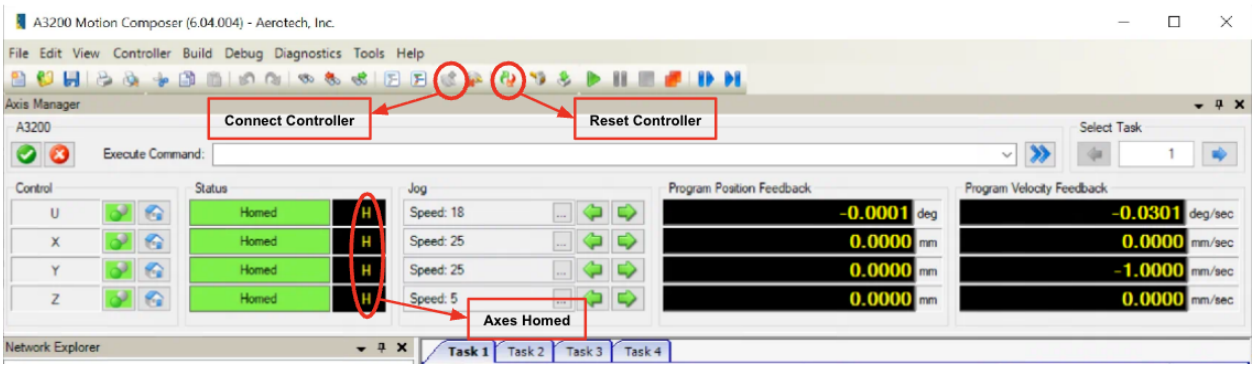


Figure 4: Motion Composer software used to operate gantry, MPG, and run G-CODE scripts for deposition.

4. Open Motion composer
5. Check that no other Gscript or Labview programs are running since they can interfere with the deposition.
6. Make sure in Motion Composer the Gantry is connected as shown in Figure 4. The elongated circle should not show VH's (virtual home) but just H's (Home). If VH is shown, the NPAQ is likely turned off.
7. Ensure the MPG program is running in Motion Composer and check that the MPG is working properly. MPGSample should be running in motion composer and dwelling on *Program Pause*.
8. Test presence of vacuum and dispenser via test programs in the same folder.
   (a) **Dispenser_Shot.vi** tests the dispenser
   (b) **manifold.vi** tests the vacuum
9. Identify parts. Handle with gloves, face mask. Check that the parts are in good condition.
10. Assign a new unique Dee number(`D_xxx`).
11. **Run control program:** `MASTER_DEE_CONSTRUCTION.vi` and provide the information required:
    (a) Part numbers.
    (b) Dee identification.
    (c) other required info
12. Mount the bottom jig plate onto the stand 1 and align it using the alignment screws (Figure 5). Repeat for second bottom jig plate onto stand 2.

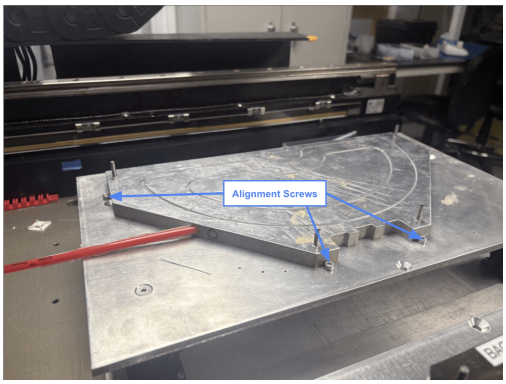


Figure 5: Jig alignment

13. Connect both labeled vacuum lines to their corresponding bottom jig.

14. Mount the CFF_A onto the bottom jig plate on Stand 1 using the jig poles and holes in the CFF_A. Repeat for CFF_B on the bottom jig on stand 2.

15. Use the neodymium magnets to secure the CFF halves onto the jig plates. Place them around the edges of the CFF halves. The magnets should lock into the correct positions.

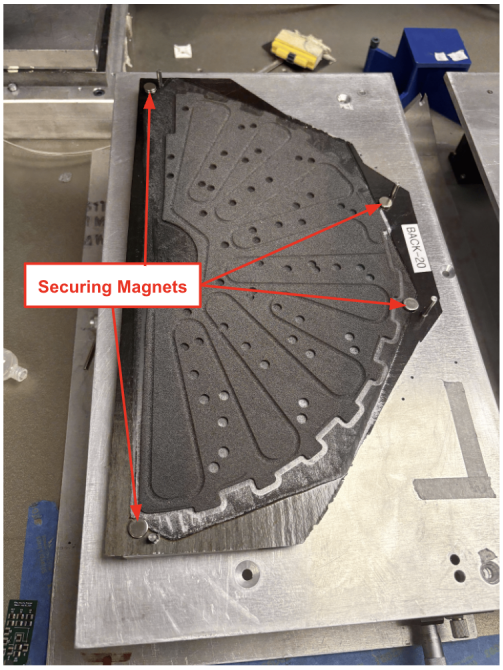


Figure 6: Neodymium magnets used to secure CFF halves onto the jig plates.

16. After person 2 comes into the clean room with both syringes and the container of doped epoxy, the control program will have moved the gantry up to the front of the table awaiting syringe installation. Screw the blue pressure cap and hose assembly onto the syringes.

17. Unscrew the blue stopper cap at the bottom of the both syringes, install the *olive green 0.063"* needle tip onto the doped-epoxy syringe AND the *light green 0.033"* needle tip onto the plain epoxy syringe and load both syringes into their designated positions on the dual syringe holder.

18. Use the dispenser pressure foot pedal and pressure splitters to bring the epoxy of each syringe to fill the tip so that there are no air bubbles in the syringe tips initially. Use a pressure $\leq$ 5.5 PSI for the plain epoxy syringe and a pressure $\leq$ 40.0 PSI for the doped epoxy syringe. Wipe off any epoxy from the tips. Select "Okay" when both syringes are installed and no air bubbles remain in the syringe tips.

19. *Initially*, the unused plain epoxy syringe should be elevated and the compressed air splitter should be open for doped epoxy (See Figure 2b).

20. The program will send the gantry home, and the user will be prompted once more before the flower pattern deposition begins. Select "Proceed" once you perform the final checks listed on the front panel prompt and the gantry will moved to the start of the flower pattern.

21. ***Ensure that you know the exact times that person 2 began mixing both epoxies* The elapsed times since the mixing of each will be used in the programs flow.**

22. Check that the olive needle tip is in the starting point for epoxy deposition as shown in figure 7. It should sit directly over the start of flower pattern line. If needle tip is not in position, adjust it using the MPG.

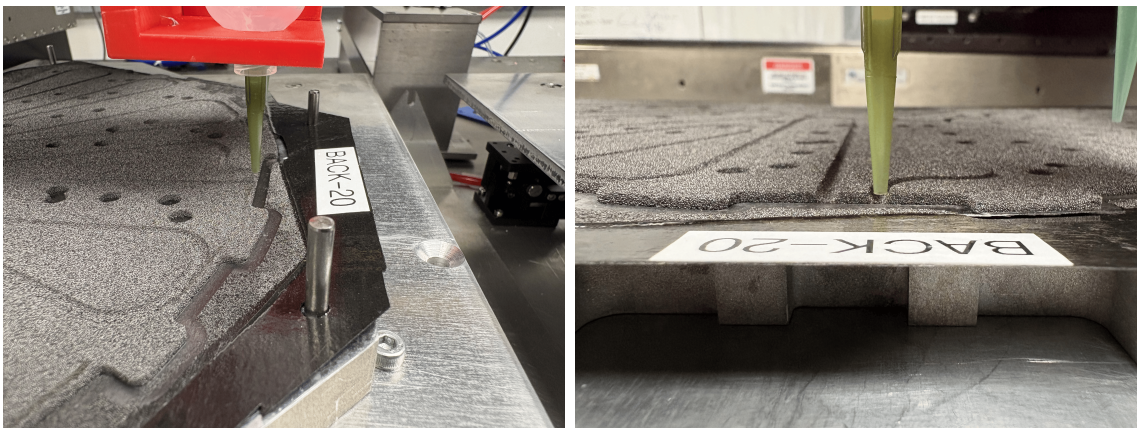


Figure 7: Needle starting position

23. [**Flower Pattern Side A**] When prompted, enter the integer amount of time that has elapsed since the doped-epoxy had been mixed. Adjust pressure to the output amount provided. Begin flower pattern doped epoxy deposition by selecting "OK" for the pressure adjust in the front panel of LabView program. Check for consistent epoxy flow and adjust the pressure if needed during the deposition.

24. [**Filling Side A**] After the flower pattern is completed, use the splitters to switch the compressed airflow to the plain epoxy syringe. Elevate the doped-epoxy syringe. When prompted enter the integer amount of elapsed time since plain epoxy was mixed, adjust the pressure accordingly. Ensure that the light green needle tip is aligned directly over the first hole in the filling deposition (adjust via MPG if not aligned) and select "OK" to proceed with Filling deposition.

25. *When each deposition ends on CFF_A, get up and check the deposition to make sure it's enough. You will be prompted on whether you would like to repeat the process of deposition for each step (Flower Pattern/Filling). If it's not enough, simply rerun the deposition ON TOP OF that deposition for CFF_A and select and YES. If it is enough, select NO. After continuing, the gantry will move to starting point for periphery deposition.*

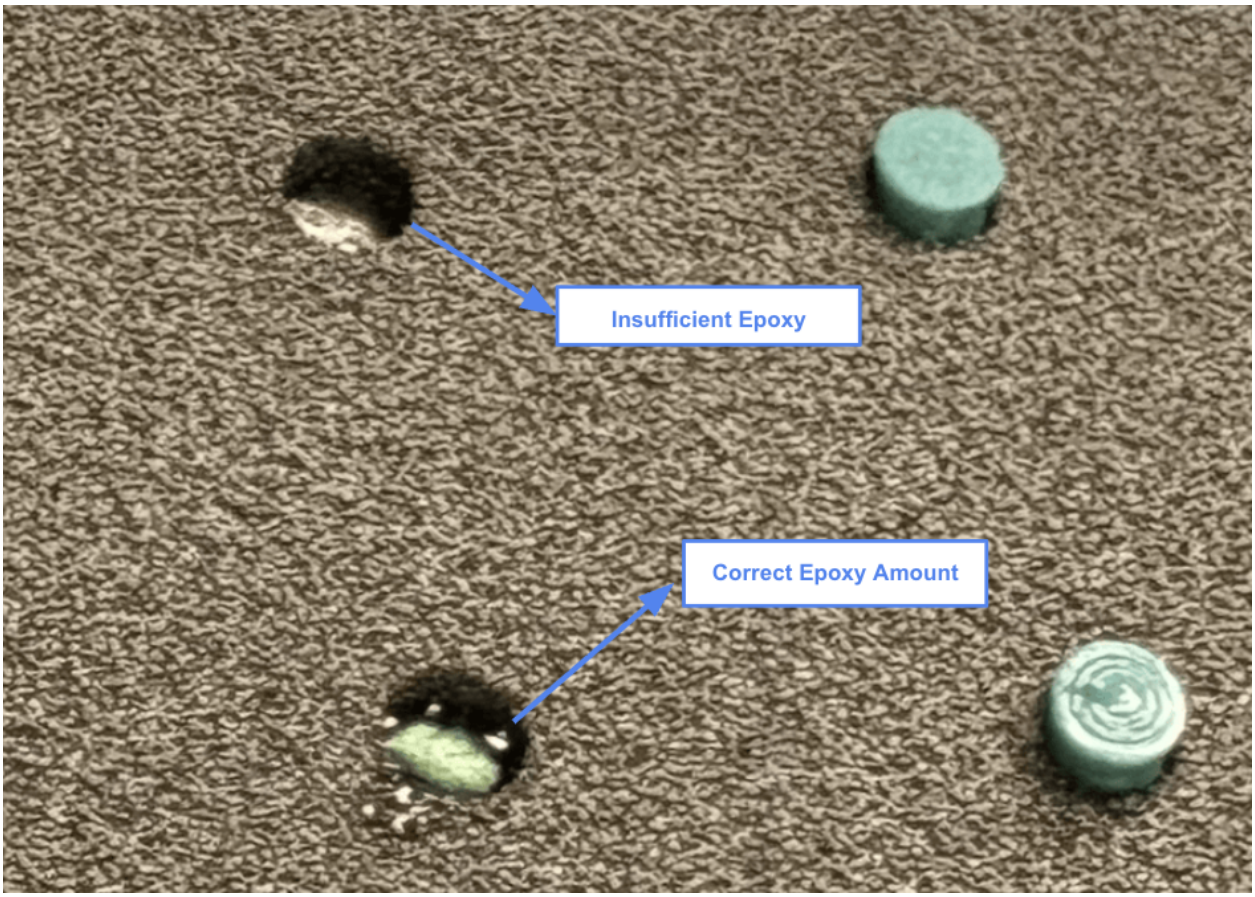


Figure 8: Plain epoxy filling hole examples

26. [**Periphery Side A**] Increase the pressure used in the prior filling deposition step by 1.0 PSI. *The start position should be just in front of the beginning of flower pattern deposition on the border encompassing the carbon foam.* Adjust position with the MPG if needed. Select "OK" to begin depositing plain epoxy around the edges of CFF_A.

27. After the periphery deposition ends, the gantry head will move out of the way of CFF_A. Remove CFF_A from the gantry table and hand to person 2

28. Before deposition for CFF_B, make sure to hand over CFF_A to Person 2, who will do the PEEK periphery, inserts, and flower pattern tubing installations.

29. Switch the compressed air splitter back to doped-epoxy and elevate the plain epoxy syringe

30. Check that the needle tip is in the starting point for flower-pattern doped epoxy deposition. Adjust position accordingly and **adjust pressure** according to the front panel pop-up.

31. [**Flower Pattern Side B**] Begin epoxy deposition by hitting OK button in the front panel of LabView program. Check for consistent epoxy flow.

32. [**Filling Side B**] Switch the splitter to plain epoxy and elevate the doped-epoxy syringe. Adjust starting position with MPG if needed. Adjust pressure according to the front panel pop-up and select "OK" to begin filling deposition.

33. *When each deposition ends on CCF_B, get up and check the deposition to make sure it's enough. You will be prompted on whether you would like to repeat the process of deposition for each step. If it's not enough, simply adjust the pressure to what you want to deposit ON TOP OF that deposition for CCF_B and select and YES. If it is enough, select NO.*

34. [**Periphery Side B**] Increase the pressure used in the prior filling deposition step by 1.0 PSI. Ensure proper starting position alignment. Select "OK" to begin depositing plain epoxy around the edges of CFF_B.

35. Remove CFF_B from the gantry table and bring to person 2, who should be finished (or almost finished) pressing the filling inserts, periphery, and evaporator tubing into CFF_A

36. Flip CFF_B so that carbon foam face towards the floor.

37. Use poles on bottom jig to stack CFF_B on top of CFF_A.

38. stack top jig plate on top of CFF_B. Make sure all of the pieces fit into place and holes are aligned when joined together.

39. Take the full stack into the oven; turn on the oven and let bake for 2 hours @ 50C, with 4h cooling. Follow instructions fixed on the oven door as shown in Figure 9.

40. Clean up the working space.

Figure 9: Oven for curing Dee after assembly

Person 2:

1. Ventilate the room with the fan
2. Put on N95 mask and gloves. Lay out paper towels over the table.
3. Materials are on the table.
4. Turn scale on in units of grams.
5. Check that centrifuge is ready to use.
6. Determine the amount of epoxy to be prepared using table in figure for the syringe with doped epoxy (Figure 10). A calculator has been set up here in case a different amount of epoxy is desired. 80g of epoxy is used as reference in the following.
7. Place the first beaker (Beaker #1) on the scale and zero the scale. Add 18.5 of Part A using one side of the Popsicle stick. Remove the beaker from the scale temporarily.
8. Place the second beaker (Beaker #2) onto the scale and zero the scale again. Add 19.2g of EA9396 Part A into beaker #2 on the scale using the same stick.
9. Using a clean popsicle stick, place 5.8g Part B into beaker #2 then mix immediately. **Take note of the time at which the mixing began.**
10. *Communicate with Person 1 the starting times of both plain and doped epoxy.*
11. Pour as the mixed plain epoxy solution from beaker #2 into the syringe and prepare it with the blue cap for the centrifuge

Epoxy calculator	A	B	Diamond	Total
35g batch ideal	8.1	2.4	24.5	35
100g batch ideal	23.1	6.9	70	100
25g batch ideal	5.8	1.7	17.5	25
Xg batch ideal	0.2	0.1	0.7	1
Xg batch ideal	0.5	0.1	1.4	2
Xg batch ideal	0.7	0.2	2.1	3
Xg batch ideal	0.9	0.3	2.8	4
Xg batch ideal	1.2	0.3	3.5	5
Xg batch ideal	1.4	0.4	4.2	6

Figure 10: Epoxy calculator table

12. Press the green start button on the centrifuge and run for 30 seconds. Ensure no bubbles remain. If so, repeat centrifuging.
13. Place beaker #1 back onto the scale
14. Zero the scale again. Add 5.5g of Part B using the other side of the stick.
15. Add 56g of diamond powder and mix until a uniform grayish-green consistency is achieved. **Also take note at this time which the doped epoxy is mixed.**
16. Prepare the syringe, don't forget to put the blue tip cap before centrifuging. Transfer the mixed epoxy into the syringe, ensuring minimal loss.
17. Centrifuge the syringe for 30 seconds at 2000 RPM. Click the green button to start and the up arrow until 2.0 shows for RPMS (2000RPM)
18. Ensure no bubbles remain inside the epoxy after centrifuging. If bubbles are present, repeat centrifuging.
19. Pour 45g of the centrifuged doped-epoxy back into beaker #1 for spreading along the carbon foam of CFF_B
20. Push the white plastic piston down onto both syringes using a sharpie/pen, making sure that the trapped air is removed.
21. Take both the plain and doped epoxy filled syringes along with beaker #1 holding the extra doped epoxy to clean room. Pass them to Person 1.
22. Get dressed to get into the clean room
23. Prepare inserts and Evaporator.

24. [**Epoxy Spreading**] While deposition for CFF_A is running, remove CFF_B from the stand and bring it to a working table. Place the flower pattern cover piece over the CFF_B flower pattern grooves. Use the extra container of doped epoxy and scrape/spread it across the carbon fiber surface of CFF_B with a paintbrush.

25. Clean out the doped-epoxy inside the filling holes (caused by the spreading) using the small spatula tool. After the holes are cleaned, remove the flower pattern cover from CFF_B.

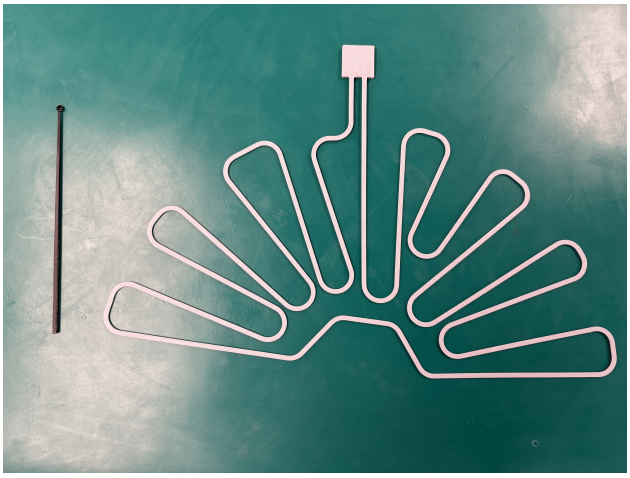


Figure 11: Flower Pattern Cover used to protect the flower pattern grooves from getting clogged with epoxy during spreading of the doped epoxy on CFF_B. Spatula/Scooping tool also included in image.

26. Return CFF_B to the stand after the depositions on CFF_A are completed, BEFORE the start of CFF_B depositions

27. Once the epoxy deposition on stand 2 starts, move bottom jig from stand 1 to working table on the left of gantry.

28. If needed, use a brush to spread the deposited plain epoxy evenly around the edges where the PEEK periphery will be placed

29. Glue the Periphery to CFF_A. Put some pressure around the periphery to ensure bonding. Use your hand/fingers.

Figure 12: Pieces used for pressing inserts and evaporator into Dee

30. Take inserts and stick them on the CFF_A carbon foam holes using the red 3D printed tools to press in (Figure 12). Be careful not to break/damage the carbon foam, although some small reshaping of the edges is acceptable as long as the fit is snug.

31. Insert the evaporator into the groove making sure that it fits. Push in with the red pressing tools.

32. Prepare top jig plate to complete the full stack.

33. Clean up the working space and all materials used. Use isopropyl alcohol to clean out the beakers and the wipes to clean all surfaces and tools.

VI. Documentation

Any special observations, e.g. damage to parts not already recorded during visual inspection, deviations from normal procedures, should be added to the report in the last step of the routine in the comments pop-up window

The following information is recorded in the report generated by the gluing software:

- Date, time (start–end) and operator name
- LabView software: version
- Id of parts used:
  - Carbon fiber-foam: S/N
  - Dee ID
  - Epoxy information
- observations/comments

Find the report at
`C:\Users\co2_ctl\Documents\Final_Dee_Assembly\Dee_Assy_Reports\Report.xls`.
Write an ELog entry and upload the report to it and to the appropriate DataBase.